

Wind farm

- **What?** Electricity-generating wind farm
- **Who?** US Windpower
- **Where and when?** US, 1980

Since 1927, the Jacobs company in the US has been creating wind turbines for generating electricity. However, these were individual units used on remote farms. It was not until 1980 that US Windpower installed 20 wind turbines on a site at Crotched Mountain, New Hampshire, making it the world's first wind farm.

Modern-day wind farm

When put underwater, the massive blades on this turbine will rotate to generate electricity.

AK1000 turbine unveiling, Scotland



Tidal energy

- **What?** Tidal power station
- **Who?** Électricité de France
- **Where and when?** France, 1966

Tide mills have existed since medieval times. Barriers let the rising tide through, then close when the tide starts going down. The captured water is then let out to power a tidal turbine. In 1996, the world's first large-scale tidal power station went into operation in France. It uses a 2,461 ft (750m) barrage across the Rance River.



Zero-carbon city

- **What?** Masdar City
- **Who?** Abu Dhabi government
- **Where and when?** UAE, 2030

Currently under construction just outside Abu Dhabi in the United Arab Emirates (UAE), Masdar City is planned to be the first city in

the world to use only renewable energy. It will be car-free, with driverless electric shuttles to move people between buildings that incorporate the latest in smart, sustainable technologies. The project began in 2006 and is expected to continue until 2030.

